

Unified QA-aware Knowledge Graph Generation Based on Multi-modal Modeling

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ABSTRACT

Understanding the long duration videos' storyline is often considered a major challenge in the field of video understanding. To promote research on understanding longer videos in the community, the deep video understanding (DVU) task is suggested for recognizing interactions at the scene level and relationships at the movie level, as well as answering questions at these two levels. In this work, we propose a unified QA-aware knowledge graph generation approach, which consists of the relation-centric graph and interaction-centric graph and demonstrates the powerful performance of multimodal pre-training models in solving such problems. Extensive validations on the HLVU dataset demonstrate the effectiveness of our proposed method.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; • **Scene understanding**; • **Knowledge representation and reasoning**;

KEYWORDS

Deep video understanding; Relationship analysis; Interaction analysis; Intent detection; Knowledge graphs

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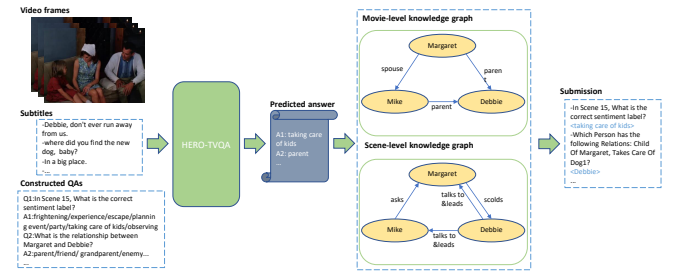


Figure 1: Pipeline of our method.

1 INTRODUCTION

With the prosperity of online media platforms, large audiences have been attracted to view abundant content with intelligent services like retrieval, recommendation and summarization tasks. Recent years have witnessed the booming of the analysis of the videos. However, due to the capacity limitation of the machine, and the complexity of modeling temporal information in long videos, it's not easy yet to parse out the holistic semantic representation of video content from longer videos, especially the dramas and movies. Along this line, Deep Video Understanding (DVU) task [3], which aims to design a unified computer system which can comprehend

events and relationships among actors and entities in an entire video automatically, is newly introduced. The DVU challenge first segments the long-form video into several scenes, and then proposes 3 types of movie-level sub-tasks and 6 types of scene-level sub-tasks for the team to solve. Different from dealing with isolated visual concepts retrieval task such as MovieQA [23] or "The Large Scale Movie Description Challenge" [18], DVU challenge requires the system to fully understand the entities and the relations among them in the video from the hierarchical perspective, which makes it more challenging.

However, the conventional video understanding framework is difficult to deal with this problem. Due to the restrictions on the scale of the training set and the long-tailed category labels, it is almost impossible to train a powerful model from scratch. Meanwhile, simply applying the pretrained ResNet backbone network to extract visual features for classification has also been proved to have poor performance. Inspired by HERO [14], which utilizes video QA as a scenario for evaluating pretrained models in downstream applications, we transform the classification task into a QA-like form to obtain answers from the model, i.e. if you want to know something, all you need is to ask the model questions, whether it's about people's relationships or interactions.

Specifically, to address the challenge and solve the above problems in deep video understanding, we propose a unified QA-aware Knowledge Graph Generation approach, which consists of Relation-centric graph and Interaction-centric graph. The key-point is to construct graphs containing the relationships and interactions of people in the video to describe its social state. And thanks to the graphs, some fine-grained semantic analysis is now available. All movie-level and scene-level sub-tasks can be resolved by our constructed knowledge graphs via the corresponding queries. To build the relation-centric as well as interaction-centric graphs, we finetune the HERO (Hierarchical Encoder for Omni-representation learning) [14] model using the frames and QA-aware subtitles in TVR [13], TVQA [12] and HLVU [3] dataset. We believe that the prior knowledge in HERO model is beneficial to the movie domain DVU challenge since it is pre-trained on the large-scale TV datasets. The description of the DVU task is shown in Figure 1.

In a word, our contribution can be summarized as proposing a novel migration and solution for video character interaction/relation-related problems. With our unified QA-aware Knowledge Graph Generation approach and several training tricks, we constructed fine-grained knowledge graphs for each video automatically and achieved remarkable results on the challenge dataset.

2 RELATED WORK

Recent years have witnessed tremendous progress in vision-language representation learning [15, 17, 19], especially in video domain [1, 11, 20]. In this paper, we encode video and corresponding subtitle via HERO [14] model, since it is pre-trained on large-scale TV datasets while the other pre-trained models focus on the general domain. Meanwhile, high level video understanding tasks involving vision and language have attracted more and more attention in the research community, which are different in evaluation forms. For instance, MovieQA [23] attempts to comprehend story both from video and text via question answering paradigm, [24] tries

to predict social relation among characters and [22] aligns book chapters with video scenes in order to better understand video contents. However, DVU challenge seeks to understand video from comprehensive perspectives, including relation prediction, graph fulfilling and retrieval-related tasks.

3 PRELIMINARY

Person search in video. Before recognizing the relations and interactions, We adopt a two-stage method to search for persons who appeared on a certain scene. First, we use YOLOv5[9] pretrained on the COCO[16] dataset to detect persons in each frame, then the detection results are passed to StrongSORT[5] based on OSNet[28] to aggregate the bounding box in adjacent video frames to generate tracklets. Regarding that the given person query images in DVU dataset is of the facial region, face detector MTCNN[26] is also applied to extract all appearing faces in tracklets. Second, we use TransReID[7] pretrained on Market1501[27] and InceptionResNetv1[21] pretrained on VGGFace2[2] for feature extraction. To effectively take full advantages of both similarities between different tracklets also similarities between tracklets and query images, we adopt PPCC[8] model to propagate person identities via both the visual and the temporal links.

Feature extraction. We refer to HERO[14] for feature extraction of video clip and its subtitle. For text embedding, we tokenize a subtitle sentence into a sequence of WordPieces[25]. Then we obtained the final representation for each sub-word token by summing up its token embedding and position embedding. For video embedding, we extract 2D and 3D visual features for each video frame by ResNet pretrained on ImageNet[4] and SlowFast[6] pretrained on Kinetics[10] respectively. We concatenate these features as visual features. Then they are fed through a fully-connected layer to be projected into the same dimensional space as token embeddings. Finally we obtain the embedding of a frame by summing up fully-connected layer outputs and position embeddings.

4 METHODOLOGY

In this section, we take the scene-level queries as an example, which contain 6 types of sub-tasks. These tasks can be described briefly as follows:

- S1:** Find Scenes based on interactions.
- S2:** Find Person based on scene number and interactions.
- S3&S4:** Find the next or previous interaction between two people.
- S5:** Find scene matching with given natural-language descriptions.
- S6:** Classify scene sentiment from a given scene.

We formulate the scene-level queries as two tasks, i.e. Building Knowledge Graphs (S1 S2 S3 S4 S6) and Video Retrieval (S5).

4.1 Pretrain and Finetune

We select HERO as our backbone model, which is pretrained on Howto100M and TV dataset. HERO uses cross-modal Transformer and captures global video context through a temporal transformer, that is suited to capture multi-character interactions and relationships. To suit the movie domain, HERO is further finetuned on TVR and TVQA dataset to obtain HERO-TVR and HERO-TVQA model respectively. However, researchers pointed out that HERO-TVQA

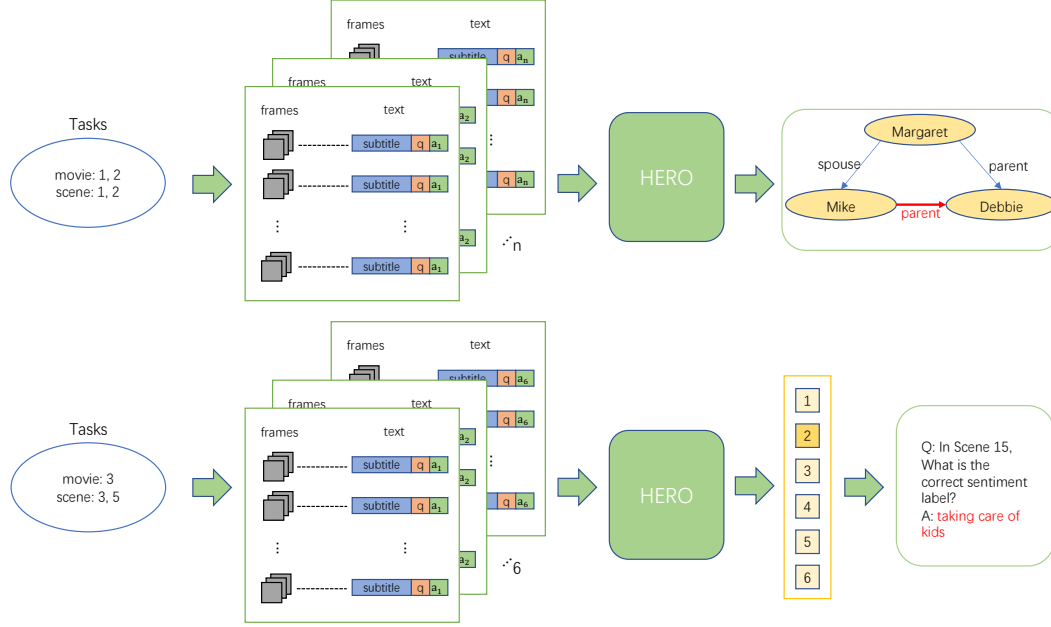


Figure 2: Solutions of building the Movie/Scene-level Knowledge Graphs.

is not performing well on DVU dataset zero-shot transferring due to distribution shift. In order to adapt the data distribution of DVU dataset, we apply further finetuning to HERO-TVQA and HERO-TVVR. For QA task, we convert the given XML format data into QA pairs as training input. For Retrieval task, we train model using the descriptions and their corresponding scenes.

4.2 Solution of Building Knowledge Graphs

Then, we will introduce how we build the scene-level knowledge graphs to answer queries, as shown in Figure 2. For each pair of characters that we are interested in, we construct a question equipped with answers to query the HERO-TVQA model about the interaction of the pair. The candidates for the question are 116 options (i.e., all interaction types) including the correct interaction between the pair. In this way, we use these multiple-choice settings to further finetune HERO-TVQA to adapt to the specific domain in the challenge dataset. During the inference process, we localize and re-identify each character or entity in scenes, for a certain pair of characters, we construct a question and answers about their interaction and feed the settings to HERO-TVQA likewise.

Inspired by the above solution, for the multiple-choice questions selected from a list of possible candidates, such as "Find next or previous interaction" and "Classify scene sentiment from a given scene", utilizing the ground truth in the scene-level interaction graph, we choose to construct the multiple-choice settings directly. To be specific, the question we encode in this step is as follows: "What is the interaction between person1 and person2?" and we equip each question with 1 correct answer and 5 random candidates. After that, we trained the HERO-TVQA using those settings. Aside from the scene-level graph, we build the movie-level graphs similarly, by establishing relationship QA pairs for different characters

in each scene and then counting the classification results in each clip to obtain the movie-level character relationships and generate the movie-level relationship graph.

4.3 Solution of Scene 1-1 Matching

Scene-level description and scene 1-1 matching is regarded as a ranking problem. during training. First, we embed scenes into vectors of $N \times L \times D$ by HERO cross-model transformer and temporal transformers, and embed queries into vectors of $N \times D$ by HERO query encoder in which N, L, D is batchsize, temporal length and embedding dimension respectively. Second, max-pooling matching confidence of $N \times N$ between each description $\mathbf{q}$ and scene $\mathbf{v}$ pair in a batch is calculated.

$$S(s_q, \mathbf{v}) = \max_{temp} \left(\frac{\mathbf{v}^{temp} \cdot \mathbf{q}}{\|\mathbf{v}^{temp}\| \|\mathbf{q}\|} \right)$$

Finally, a hinge loss $\max(0, \delta + S_{neg} - S_{pos})$ is backpropagated to optimize the model. Both query negative samples and scene negative samples are considered.

During inference stage, we normalize the scores of all scenes for each description. Then Hungarian algorithm is applied to find a proper matching strategy. The algorithm can solve bipartite graph matching problem with global optimum.

4.4 Solution of Queries

After constructing the movie-level knowledge graphs and scene-level graphs through our proposed method, we are able to provide answers to most of queries. For the task of "Relations between characters", all valid paths from one entity to another in the constructed graph are submitted. For the task of "Fill in the graph space" in both

Movie and scene level, each triple in the list of person attributes, interactions, relationships, events or actions to and from other people, will be transformed into the form of QA independently. The results will be a ranked list of answers ordered by occurrence frequency. For the task of “Question Answering”, we found the type of questions can be divided into ‘Integer: Unknown’, ‘Relation: Unknown’ and ‘Entity: Unknown’, and six candidate answers are given. We still model the questions of ‘Relation’ and ‘Entity’ as the QA problem, which is intuitive. Yet, we use the constructed KG to count the number of relations to answer questions of ‘Integer’. For the task of “Find the Unique Scene”, we traverse all scene graphs and count the number of interactions covered and uncovered by the given list of interactions. The scene that have more covered interactions and less uncovered interactions will be ranked higher. For the task of “Find next or previous interaction”, we traverse all the candidates to find a best interaction. The solution for “Find the 1-1 relationship between scenes and natural language descriptions” are described in Section 4.3. For the task of “Classify scene sentiment from a given scene”, we also model it as a QA problem. However, instead of asking about the relationship/interaction between entities, the question is to ask “What is the sentiment of this scene?”. In this way, we can directly answer this query in the inference phase.

5 EXPERIMENTS

5.1 Dataset and Experimental Settings

The full Deep Video Understanding set contains 14 open source movies for training which lasts about 16 hours in total, and 6 movies are added as testing dataset. During the experiments, to evaluate the performance of the QA task and building graph, we choose **Acc** and **Hit@n** as the metrics.

5.2 Evaluation and Results

The experimental results on validation set are listed in Table 1 and Table 2. Specifically, based on the experimental results, there are several noteworthy findings:

For the task of building graphs, the utilization of full interaction/relationship labels in constructing the questions and answers leads to a huge challenge to the model naturally. However, we find that our model has a good performance in revealing the people’s interactions, the reason may be that our model makes full and reasonable use of multi-modal information. While using video frames for modeling the character interaction from the visual perspective, the subtitles can enhance the semantic representation additionally. Thus, an accuracy rate of 19.87 was achieved in the answering of the interaction question equipped with 116 candidates. What’s more, as shown in Table 1, the HERO-TVQA model trained individually for the multiple-choice questions achieves the accuracy of 26.12 and 54.06 at movie-level and scene-level, respectively, which is a pretty example of the advantage of the setting.

For the scene-description 1-1 matching task, we follow the latest year DVU challenge data splitting. There are 351 pairs in training set and 177 pairs in validation set. We believe two hyper-parameter settings are important: max frame length of video and negative sampling strategy. The model may be sensitive to max frame length of video because of the max-pooling rating design. The experiments in table 2 show that 120 frame length is better than 140 frame length

setting. Besides, negative sampling strategy provides more diverse contrastive option for the model training than using all negative samples. The results in table 2 also support our assertion. Finally, we achieve 26.55 Recall score for retrieving 1 from 177 scenes.

Moreover, the final scores on all scene-level queries of the testing dataset are shown in Table 3. All the movies are newly offered in this year’s testing dataset and the result is evaluated on the ground truth owned by the organizers.

Table 1: Building Graphs

Task	Question Answer(6QA)		Building Graphs(68QA/116QA)		
	Acc	Hit@3	Acc	Hit@10	Hit@50
Movie-Level	26.12	60.18	2.95	25.05	85.08
Scene-Level	54.06	83.51	19.87	43.12	75.43

Table 2: Scene Matching

Max frame length	Negative sampling	R@1	R@5	R@10
140	sample	23.73	50.85	59.89
120	sample	26.55	50.28	66.10
120	full	24.86	49.15	63.84

Table 3: Evaluation(%) of Queries on Testing Dataset

Movie	S1	S2	S3	S4	S5	S6
Chained for life	15.00	71.00	50.00	33.00	80.00	0.00
Liberty kid	10.00	36.00	25.00	0.00	80.00	0.00
Like me	5.00	21.00	0.00	0.00	40.00	17.00
Little rock	0.00	0.00	0.00	50.00	60.00	50.00
Losing ground	5.00	19.00	25.00	75.00	80.00	33.00
Calloused hands	11.00	31.00	25.00	0.00	50.00	17.00
Mean	7.00	30.00	21.00	26.00	65.00	19.00

6 CONCLUSIONS

In this paper, we propose a unified QA-aware Knowledge Graph Generation approach based on the pre-trained multimodal models to build relation-centric graphs and interaction-centric graphs for deep video understanding tasks. The models jointly use the subtitles and video frames to analyze the relationships and interactions among entities in movies. The result on the HLVD dataset validated the effectiveness of our solution to the related tasks.

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